

MAT 426: Topics in Linear Algebra

Exam 1

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Solutions

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1. (10 points) Let

$$A = \begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 7 \\ 8 & 7 & 9 & 19 \\ 6 & 7 & 9 & 20 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 4 \end{bmatrix}$$

(a) Find the LU factorization $A = LU$ where L is unit lower triangular.

$$\begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 7 \\ 8 & 7 & 9 & 19 \\ 6 & 7 & 9 & 20 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 4R_1 \\ R_4 \leftarrow R_4 - 3R_1}} \begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & 5 & 7 \\ 0 & 4 & 6 & 11 \end{bmatrix}$$

$$\xrightarrow{\substack{R_3 \leftarrow R_3 - 3R_2 \\ R_4 \leftarrow R_4 - 4R_2}} \begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 7 \end{bmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix} = U$$

$$\begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 7 \\ 8 & 7 & 9 & 19 \\ 6 & 7 & 9 & 20 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 \\ 3 & 4 & 1 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}}_U$$

(b) Use the factorization to solve the system $A\mathbf{x} = \mathbf{b}$.

$$L\bar{\mathbf{y}} = \bar{\mathbf{b}}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 \\ 3 & 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 4 \end{bmatrix}$$

$$\begin{aligned} y_1 &= 1 \\ 2y_1 + y_2 &= 3 \Rightarrow y_2 = 1 \\ 4y_1 + 3y_2 + y_3 &= 5 \Rightarrow y_3 = -2 \\ 3y_1 + 4y_2 + y_3 + y_4 &= 4 \Rightarrow y_4 = -1 \end{aligned}$$

$$U\bar{\mathbf{x}} = \bar{\mathbf{y}}$$

$$\begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \\ -1 \end{bmatrix}$$

$$\begin{aligned} 3x_4 &= -1 \Rightarrow x_4 = -1/3 \\ 2x_3 + 4x_4 &= -2 \Rightarrow x_3 = -1/3 \\ x_2 + x_3 + x_4 &= 1 \Rightarrow x_2 = 5/3 \\ 2x_1 + x_2 + x_3 + 3x_4 &= 1 \Rightarrow x_1 = 1/3 \end{aligned}$$

$$\bar{\mathbf{x}} = \begin{bmatrix} 1/3 \\ 5/3 \\ -1/3 \\ -1/3 \end{bmatrix}$$

2. (8 points) Answer the following by circling either TRUE or FALSE.

- True ☒ False If a matrix A has an LU factorization, then the factorization is unique.
- ☒ True False Finding the LU factorization of an $n \times n$ matrix A requires $\frac{2}{3}n^3 + \mathcal{O}(n^2)$ flops.
- ☒ True False Gaussian elimination may fail if a pivot is zero unless row swaps are allowed.
- True ☒ False When implemented in a computer, Gaussian elimination without pivoting will always produce matrices LU with $A = LU$ as long as a zero pivot is not encountered.
- ☒ True False If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
- ☒ True False If U is an orthogonal matrix, then $U^{-1} = U^T$.
- True ☒ False If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u}$ and $\mathbf{v}$ must both be nonzero.
- True ☒ False The projection of a vector onto another vector is always orthogonal to that vector.

3. (3 points) Complete the following statements.

- (a) The product of a permutation matrix with itself produces an identity matrix.
- (b) The product of two permutation matrices is a permutation matrix.
- (c) Multiplying an elementary matrix M on both the left and right by a permutation matrix P produces a elementary matrix.

4. (6 points) Gaussian elimination with partial pivoting directly yields the factorization $MA = U$ where M is a permuted elementary matrix and U is an upper triangular matrix. For a 3×3 matrix A , show how M is formed and how the $PA = LU$ factorization is extracted.

$$\begin{aligned} U &= M_2 P_2 M_1 P_1 A \Rightarrow M = M_2 P_2 M_1 P_1 \\ &= M_2 P_2 \cdot M_1 P_2 P_1 A \\ &= M_2 (P_2 M_1 P_2) \cdot (P_2 P_1) A \\ &= M_2' M_1' P A \Rightarrow P = P_2 P_1 \end{aligned}$$

$$L = (M_1')^{-1} (M_2')^{-1}.$$

5. (12 points) Suppose A , L and U are $n \times n$ matrices.

(a) Show that solving an upper triangular system $U\mathbf{x} = \mathbf{y}$ requires n^2 floating point operations (flops).

$$\begin{aligned} \sum_{j=1}^n \underbrace{(j-1)}_{\text{multiplications}} + \underbrace{(j-1)}_{\text{subtractions}} + \underbrace{1}_{\text{division}} &= \sum_{j=1}^n 2j-1 \\ &= 2 \frac{n(n+1)}{2} - n \\ &= n^2 + n - n \\ &= n^2 \end{aligned}$$

(b) How many flops are required to solve a unit lower triangular system $L\mathbf{y} = \mathbf{b}$?

$$n^2 - n \quad (\text{no divisions})$$

(c) Performing row operations to row reduce $[A|I]$ to $[I|A^{-1}]$ is equivalent to solving n linear systems $A\mathbf{x} = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th column of the identity matrix I . How many floating point operations are needed to solve these n linear systems using the LU decomposition of A ?

$$[n^2 + (n^2 - n)]n = 2n^3 - n^2$$

(d) How many floating point operations are required to compute A^{-1} ?

$$\frac{8}{3}n^3 + O(n^2) + 2n^3 - n^2 = \frac{8}{3}n^3 + O(n^2)$$

(e) How many floating point operations are needed to multiply $A^{-1}\mathbf{b}$?

$$\left(\underbrace{n+n}_{\text{multiplications, additions}} \right) (\# \text{ rows}) = 2n \cdot n = 2n^2$$

(f) Show that $\frac{8}{3}n^3 + O(n^2)$ floating point operations are needed to solve $A\mathbf{x} = \mathbf{b}$ by finding the inverse of A using the method outlined above.

$$(d) + (e) = \frac{8}{3}n^3 + O(n^2) + 2n^2 = \frac{8}{3}n^3 + O(n^2)$$

6. (12 points) Let

$$A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix}$$

- (a) Use Gaussian elimination with partial pivoting to find the $PA = LU$ decomposition of A . Clearly state the matrices P , L , and U .

$$\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix}$$

$$P_1 A = \begin{bmatrix} 2 & 4 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 0 & 2 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix} \quad \begin{array}{l} R_2 \leftarrow R_2 - \frac{1}{2}R_1 \\ R_4 \leftarrow R_4 - \frac{1}{2}R_1 \end{array} \quad \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$P_2 P_1 A = \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 2 & 1 & 3 \\ 0 & -1 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \begin{array}{l} R_3 \leftarrow R_3 + \frac{2}{3}R_2 \\ R_4 \leftarrow R_4 - \frac{1}{3}R_2 \end{array} \quad \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 2 & \frac{8}{3} \\ 0 & 0 & -1 & \frac{2}{3} \end{bmatrix} \quad \begin{array}{l} R_4 \leftarrow R_4 + \frac{1}{2}R_3 \end{array} \quad \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 2 & \frac{8}{3} \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$PA = LU$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & -\frac{2}{3} & 1 & 0 \\ \frac{1}{2} & \frac{1}{3} & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 2 & \frac{8}{3} \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

- (b) Suppose GEPP is implemented in a computer algorithm. State the matrix A and vector r that would be returned by this algorithm.

$$\bar{r} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 4 & 1 & 3 \\ \frac{1}{2} & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -\frac{2}{3} & 2 & \frac{8}{3} \\ \frac{1}{2} & \frac{1}{3} & -\frac{1}{2} & 2 \end{bmatrix}$$

7. (4 points) Define the following.

(a) The inner product of two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is $\bar{\mathbf{u}} \cdot \bar{\mathbf{v}} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

(b) Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if and only if $\bar{\mathbf{u}} \cdot \bar{\mathbf{v}} = \langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle = 0$

8. (11 points) Let $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}$. Compute the following:

(a) $\mathbf{u} \cdot \mathbf{v} = (1)(0) + (3)(-5) + (-2)(4) = -15 - 8 = -23$

(b) $\|\mathbf{u} - \mathbf{v}\| = \sqrt{(1-0)^2 + (3-(-5))^2 + (-2-4)^2} = \sqrt{1 + 64 + 36} = \sqrt{101}$

(c) Write $\mathbf{v}$ as the sum of a vector in $\text{Span}\{\mathbf{u}\}$ and a vector orthogonal to $\mathbf{u}$

$$\text{proj}_{\bar{\mathbf{u}}} \bar{\mathbf{v}} = \frac{\bar{\mathbf{u}} \cdot \bar{\mathbf{v}}}{\bar{\mathbf{u}} \cdot \bar{\mathbf{u}}} \bar{\mathbf{u}} = \frac{-23}{14} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -23/14 \\ -69/14 \\ 46/14 \end{bmatrix}$$

$$\text{perp}_{\bar{\mathbf{u}}} \bar{\mathbf{v}} = \bar{\mathbf{v}} - \text{proj}_{\bar{\mathbf{u}}} \bar{\mathbf{v}} = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix} - \begin{bmatrix} -23/14 \\ -69/14 \\ 46/14 \end{bmatrix} = \begin{bmatrix} 23/14 \\ -1/14 \\ 10/14 \end{bmatrix}$$

$$\bar{\mathbf{v}} = \begin{bmatrix} -23/14 \\ -69/14 \\ 46/14 \end{bmatrix} + \begin{bmatrix} 23/14 \\ -1/14 \\ 10/14 \end{bmatrix}$$

9. (8 points) Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. The Cauchy-Schwarz inequality states that $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$. Use the Cauchy-Schwarz inequality to prove that $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

$$\begin{aligned} \|\bar{\mathbf{u}} + \bar{\mathbf{v}}\|^2 &= \langle \bar{\mathbf{u}} + \bar{\mathbf{v}}, \bar{\mathbf{u}} + \bar{\mathbf{v}} \rangle \\ &= \langle \bar{\mathbf{u}}, \bar{\mathbf{u}} \rangle + 2\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle + \langle \bar{\mathbf{v}}, \bar{\mathbf{v}} \rangle \\ &= \|\bar{\mathbf{u}}\|^2 + 2\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle + \|\bar{\mathbf{v}}\|^2 \\ &\leq \|\bar{\mathbf{u}}\|^2 + 2|\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle| + \|\bar{\mathbf{v}}\|^2 \\ &\leq \|\bar{\mathbf{u}}\|^2 + 2\|\bar{\mathbf{u}}\| \cdot \|\bar{\mathbf{v}}\| + \|\bar{\mathbf{v}}\|^2 \\ &= (\|\bar{\mathbf{u}}\| + \|\bar{\mathbf{v}}\|)^2 \end{aligned}$$

$$\Rightarrow \|\bar{\mathbf{u}} + \bar{\mathbf{v}}\| \leq \|\bar{\mathbf{u}}\| + \|\bar{\mathbf{v}}\|$$

10. (12 points) Let $S = \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \mathbf{u}_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ -2 \end{bmatrix} \right\}$ and let $\mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$.

(a) Show that S is an orthogonal set.

$$\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_2 = 0+0+0+0=0 \quad \bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_3 = 0+1+(-1)+0=0$$

$$\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_3 = 0+0+0+0=0 \quad \bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_4 = 0+1+(-1)+0=0$$

$$\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_4 = 0+0+0+0=0 \quad \bar{\mathbf{u}}_3 \cdot \bar{\mathbf{u}}_4 = 0+1+1+(-2)=0$$

$\Rightarrow S$ is an orthogonal set

(b) Explain why S forms an orthogonal basis for $\mathbb{R}^4$.

An orthogonal set of vectors must be linearly independent and since we have four vectors in $\mathbb{R}^4$ they must span $\mathbb{R}^4$; i.e. S is an orthogonal basis for $\mathbb{R}^4$.

(c) Write $\mathbf{y}$ as a linear combination of the vectors in S .

$$\begin{aligned} \hat{\mathbf{y}} &= \frac{\langle \bar{\mathbf{y}}, \bar{\mathbf{u}}_1 \rangle}{\langle \bar{\mathbf{u}}_1, \bar{\mathbf{u}}_1 \rangle} \bar{\mathbf{u}}_1 + \frac{\langle \bar{\mathbf{y}}, \bar{\mathbf{u}}_2 \rangle}{\langle \bar{\mathbf{u}}_2, \bar{\mathbf{u}}_2 \rangle} \bar{\mathbf{u}}_2 + \frac{\langle \bar{\mathbf{y}}, \bar{\mathbf{u}}_3 \rangle}{\langle \bar{\mathbf{u}}_3, \bar{\mathbf{u}}_3 \rangle} \bar{\mathbf{u}}_3 + \frac{\langle \bar{\mathbf{y}}, \bar{\mathbf{u}}_4 \rangle}{\langle \bar{\mathbf{u}}_4, \bar{\mathbf{u}}_4 \rangle} \bar{\mathbf{u}}_4 \\ &= \frac{3}{1} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \frac{3}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \frac{5}{3} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix} - \frac{7}{6} \begin{bmatrix} 0 \\ 1 \\ -1 \\ -2 \end{bmatrix} \end{aligned}$$

(d) Is S an orthonormal set? If not, find an orthonormal set S' such that $\text{Span}(S') = \text{Span}(S)$.

No, $\|\bar{\mathbf{u}}_2\| \neq 1$

$$S' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \begin{bmatrix} 0 \\ 1/\sqrt{6} \\ -1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix} \right\}$$

11. (8 points) Prove that an orthogonal set of nonzero vectors is linearly independent.

Suppose $S = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_p\}$ is an orthogonal set of nonzero vectors. and consider $c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_p\vec{u}_p = \vec{0}$. Taking the dot product of this with $\vec{u}_j$ gives

$$(c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_p\vec{u}_p) \cdot \vec{u}_j = \vec{0} \cdot \vec{u}_j = 0$$

$$c_1(\vec{u}_1 \cdot \vec{u}_j) + \dots + c_j(\vec{u}_j \cdot \vec{u}_j) + \dots + c_p(\vec{u}_p \cdot \vec{u}_j) = 0$$

$$c_j(\vec{u}_j \cdot \vec{u}_j) = 0$$

Since the vectors are orthogonal. Since $\vec{u}_j \neq \vec{0}$ $\vec{u}_j \cdot \vec{u}_j = \|\vec{u}_j\|^2 > 0$. Thus, $c_j = 0$.

This holds for all j . Thus the only solution to $c_1\vec{u}_1 + \dots + c_p\vec{u}_p = \vec{0}$ is the trivial solution $c_1 = \dots = c_p = 0$; that is, the vectors are linearly independent.

12. (6 points) Consider the matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{3}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & -\frac{\sqrt{3}}{4} \\ \frac{1}{2} & -\frac{1}{2\sqrt{3}} & -\frac{1}{4} \\ \frac{1}{2} & -\frac{1}{\sqrt{3}} & \frac{\sqrt{3}}{4} \end{bmatrix}$.

(a) Determine whether or not A is an orthonormal matrix.

$$\begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{4} \\ -\frac{1}{4} \\ \frac{\sqrt{3}}{4} \end{bmatrix} = \frac{1}{4} - \frac{\sqrt{3}}{8} - \frac{1}{8} + \frac{\sqrt{3}}{8} = \frac{1}{8} \neq 0$$

Thus A is not orthogonal and cannot be orthonormal.

or

$$\left\| \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \right\| = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}} = 1 \quad \left\| \begin{bmatrix} \frac{\sqrt{3}}{4} \\ \frac{1}{2\sqrt{3}} \\ -\frac{1}{4\sqrt{3}} \\ \frac{1}{4\sqrt{3}} \end{bmatrix} \right\| = \sqrt{\frac{1}{3} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}} = \sqrt{\frac{2}{3}} \neq 1$$

(b) Is A an orthogonal matrix? Explain.

See (a).